

Configuring a VPN Client for Secure File Transfers (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 09

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Time on Task:

2 hours, 16 minutes

Progress:

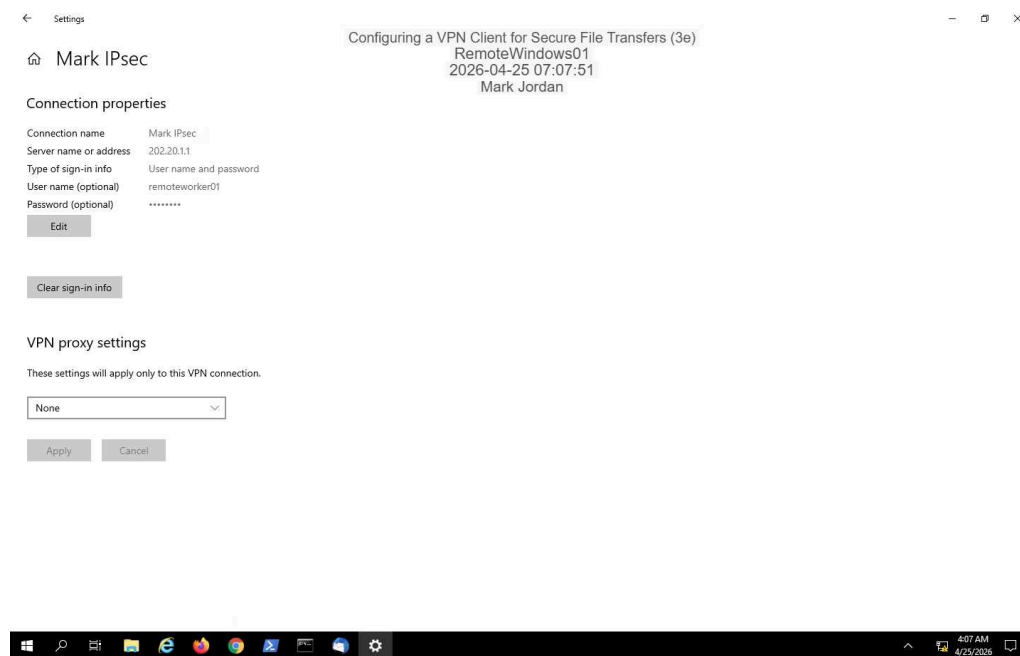
100%

Report Generated: Saturday, April 25, 2026 at 9:11 AM

Section 1: Hands-On Demonstration

Part 1: Configure a Windows VPN Client

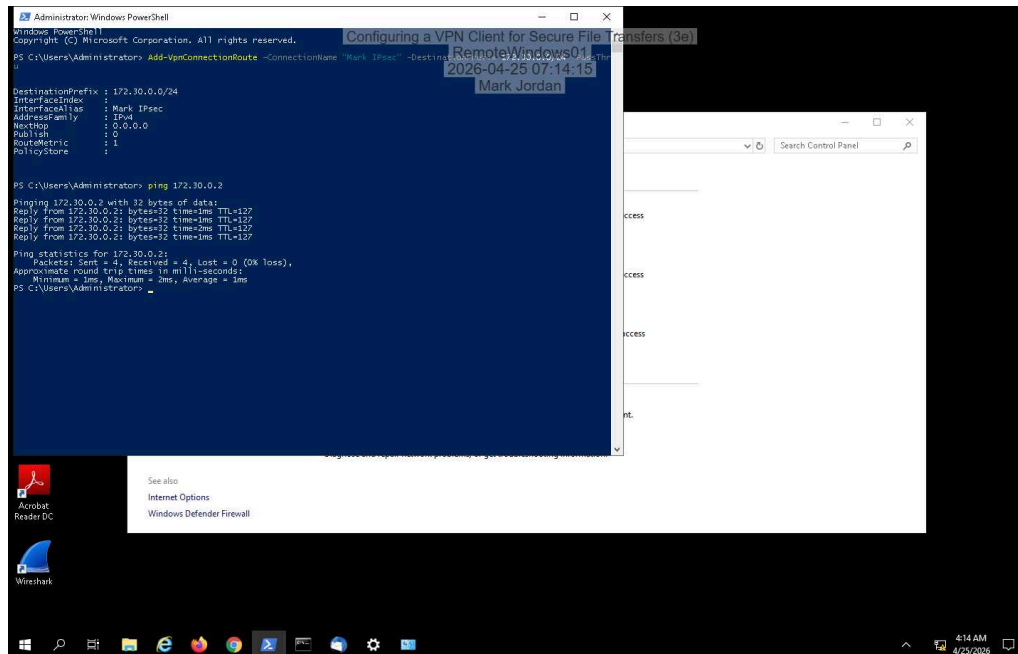
30. Make a screen capture showing the VPN connection properties.



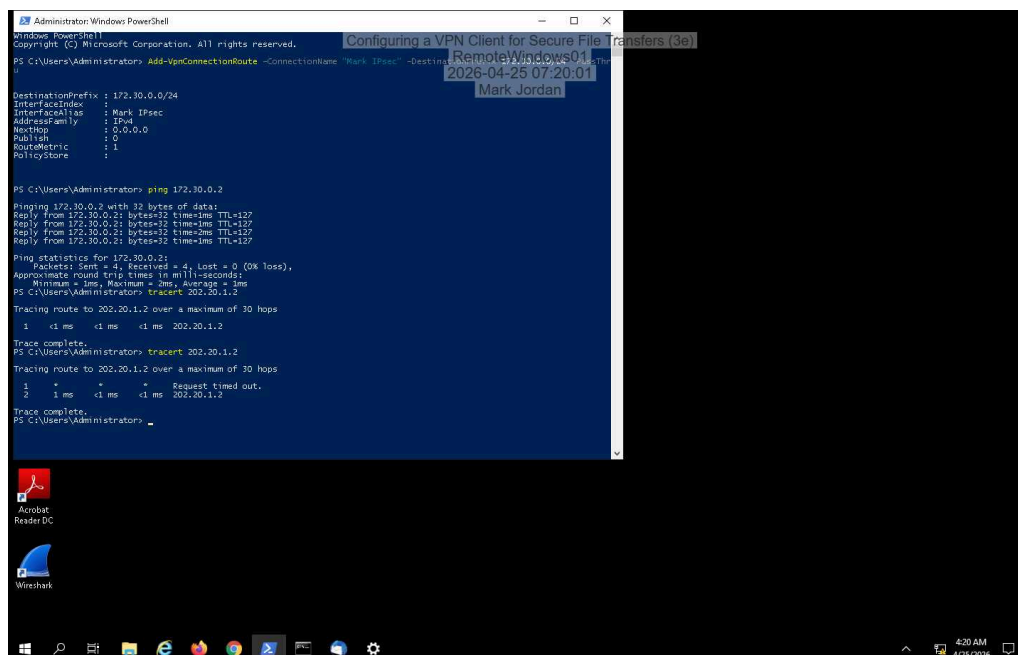
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54. Make a screen capture showing the successful ping response.

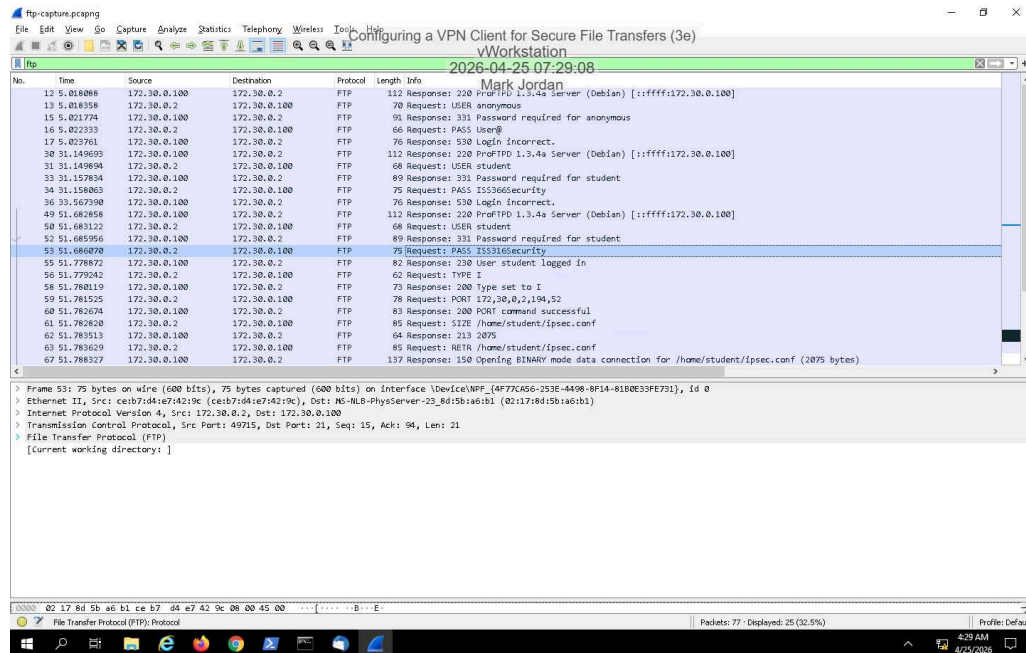


72. Make a screen capture showing your new tracert results.

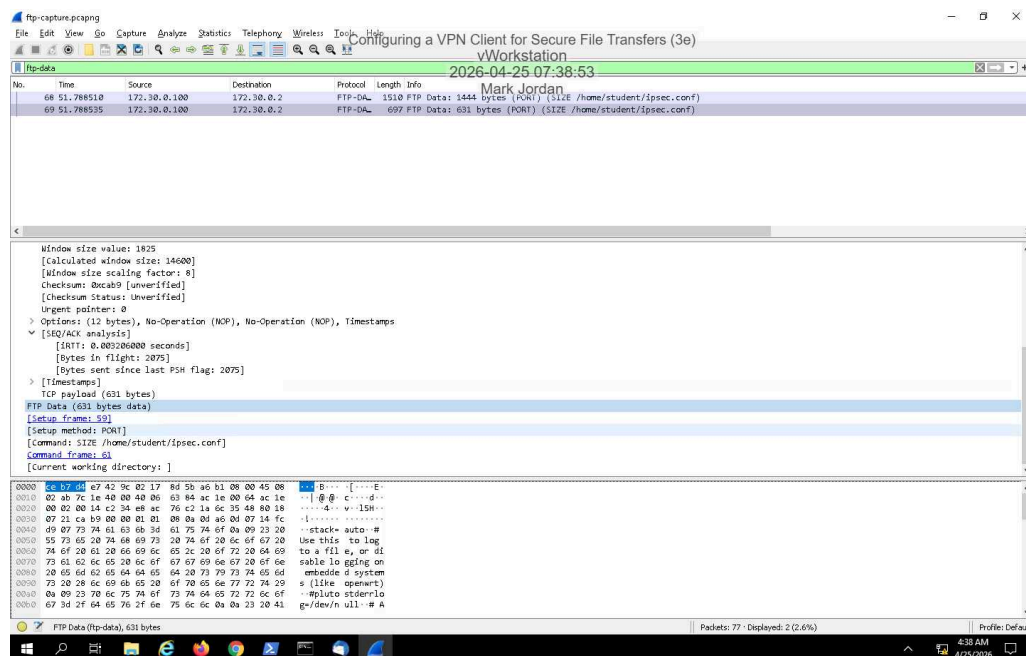


Part 2: Compare Secure and Non-Secure File Transfers in Wireshark

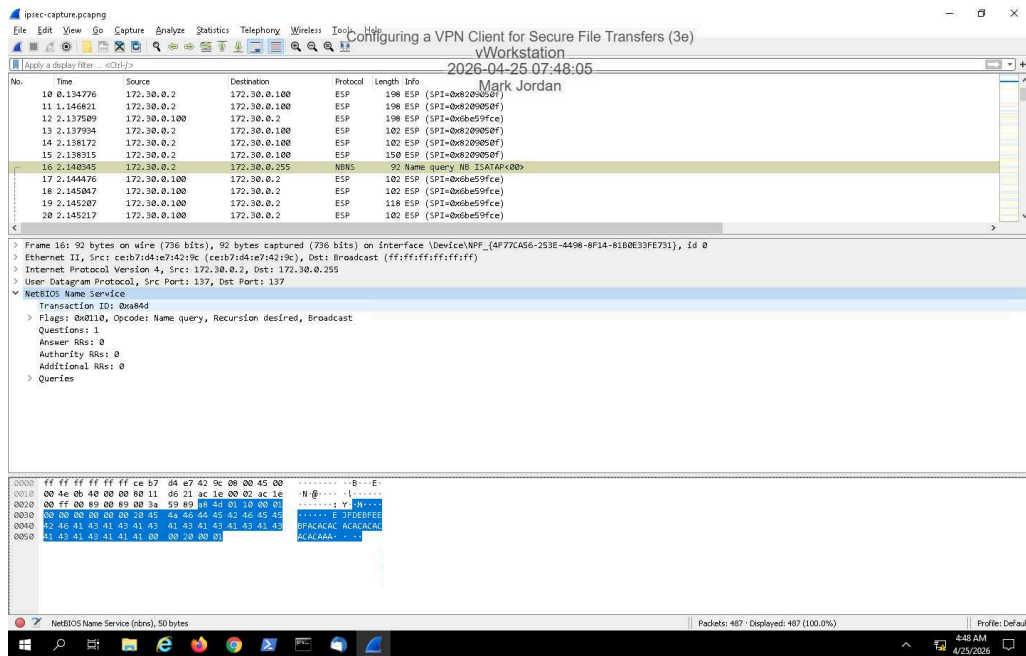
12. Make a screen capture showing the packet that carries the correct password.



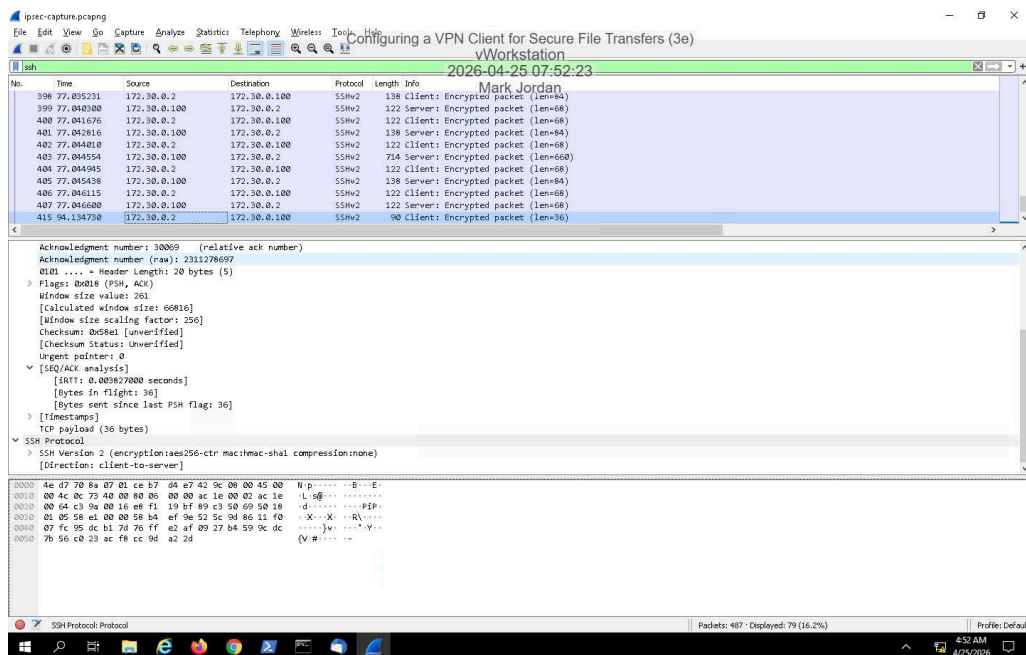
28. Make a screen capture showing the Wireshark window and the packet bytes pane for Packet 69.



44. Make a screen capture showing the packet details pane for packet 16.



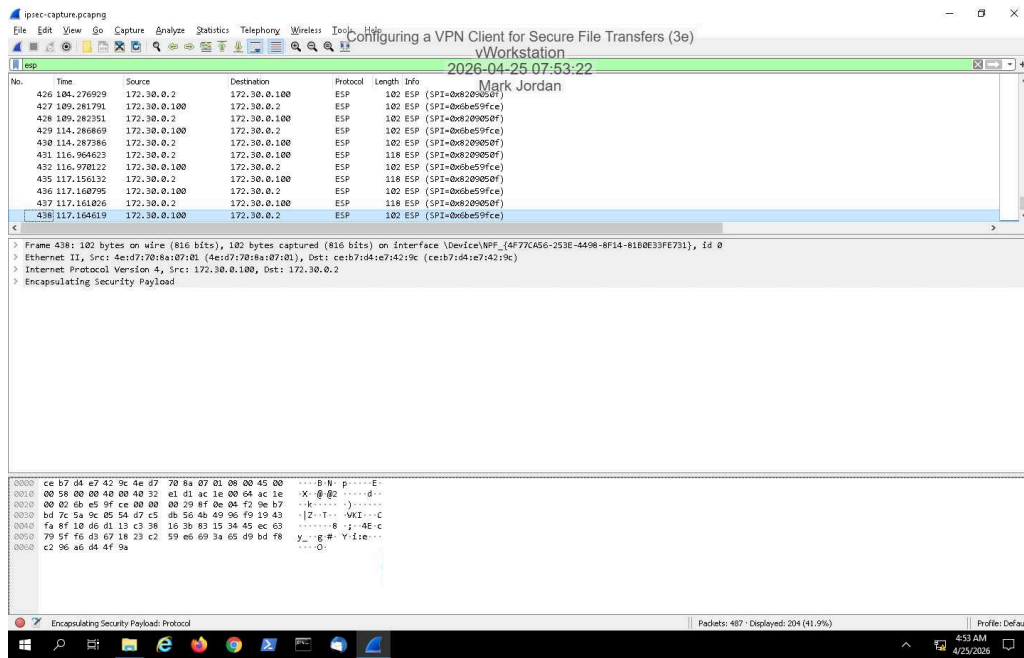
49. Make a screen capture showing the last SSHv2 packet in the SSH file transfer.



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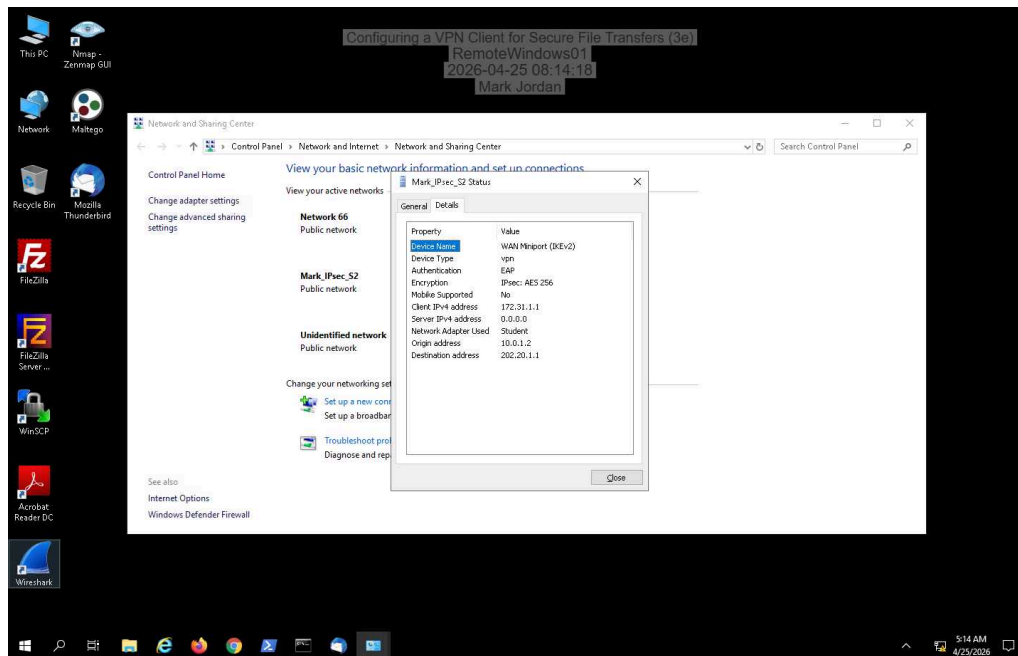
51. Make a screen capture showing the last packets in the ESP exchange.



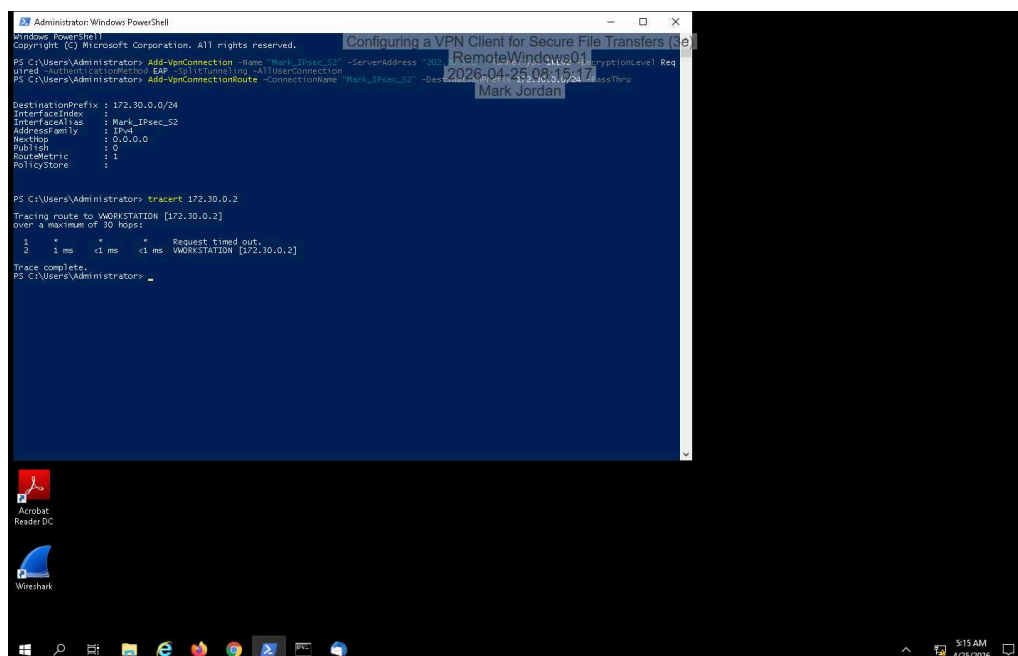
Section 2: Applied Learning

Part 1: Configure a Windows VPN Client

19. Make a screen capture showing the IPsec VPN connection encrypted with AES 256.



23. Make a screen capture showing your successful tracert to the remote machine.

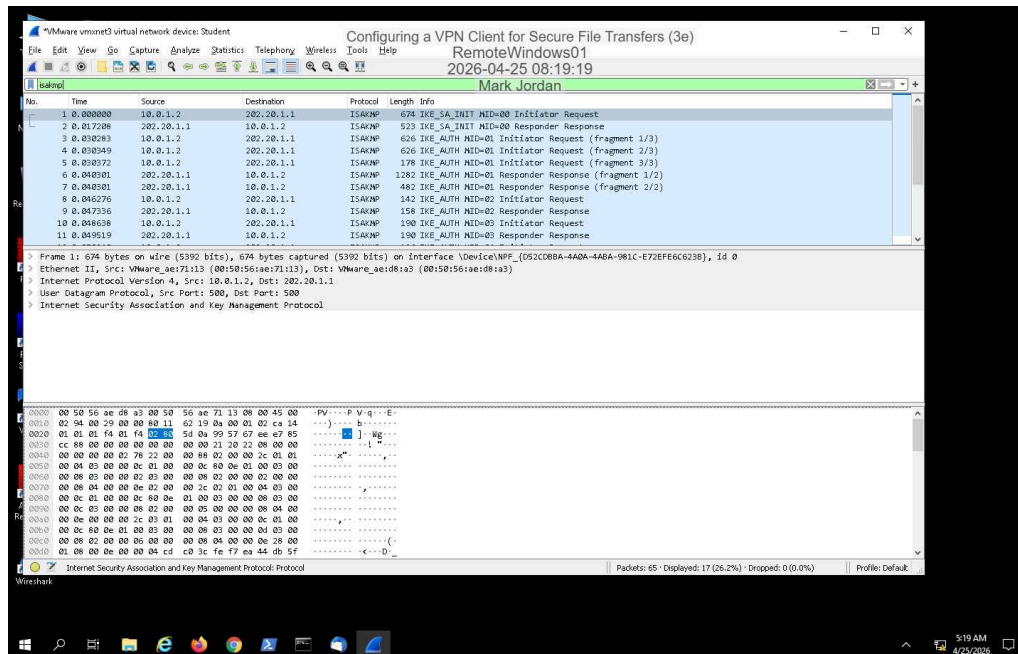


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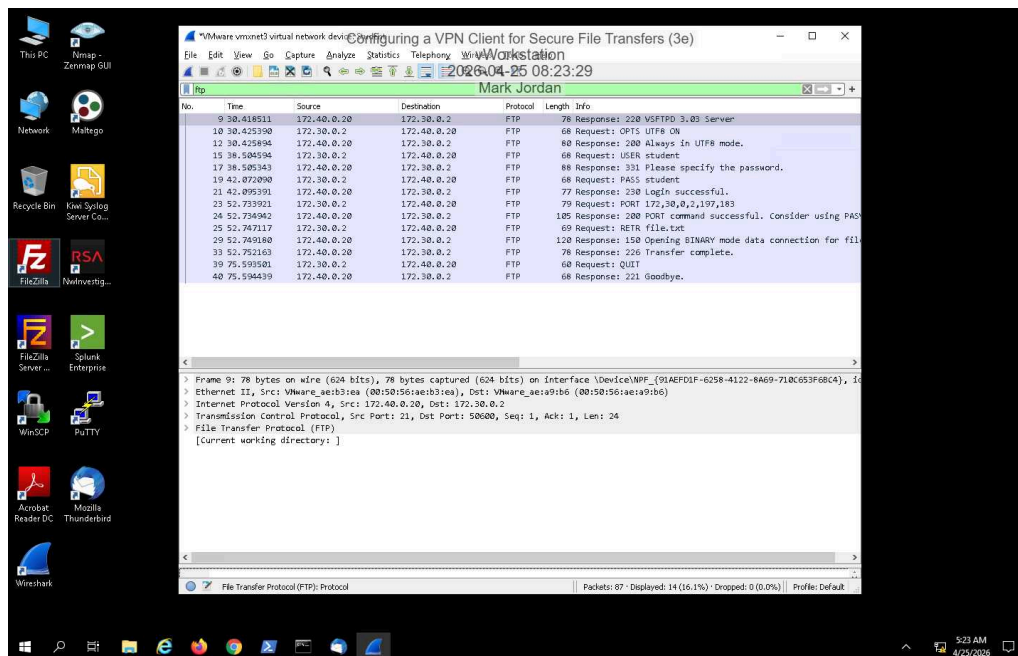
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Part 2: Compare Secure and Non-Secure File Transfers in Wireshark

7. Make a screen capture showing the IKE_SA_INIT and IKE_AUTH packets.



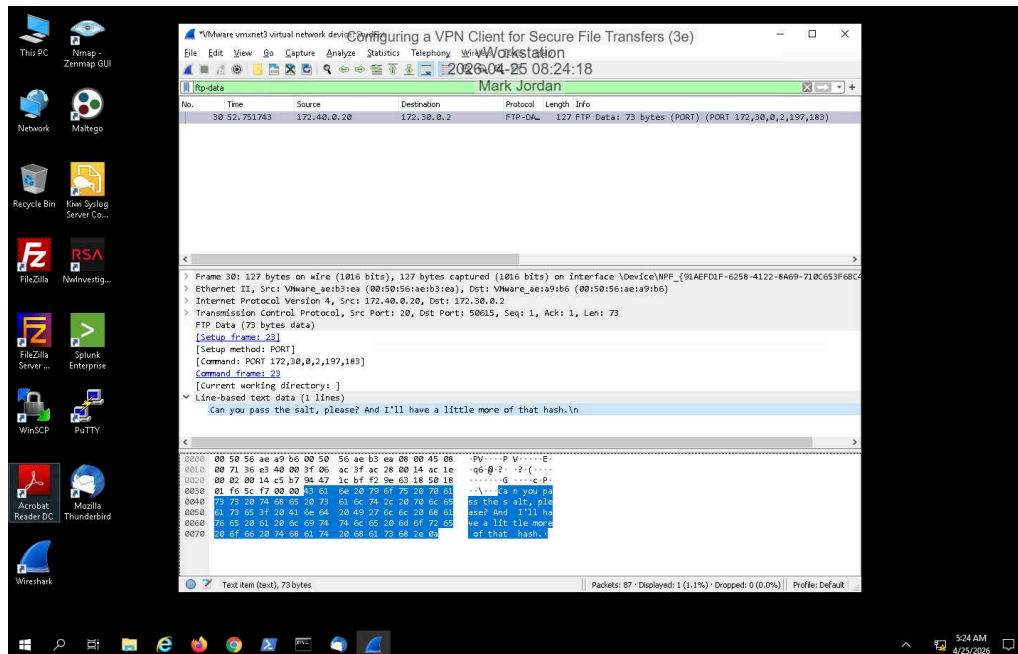
22. Make a screen capture showing the filtered FTP packets in your capture file.



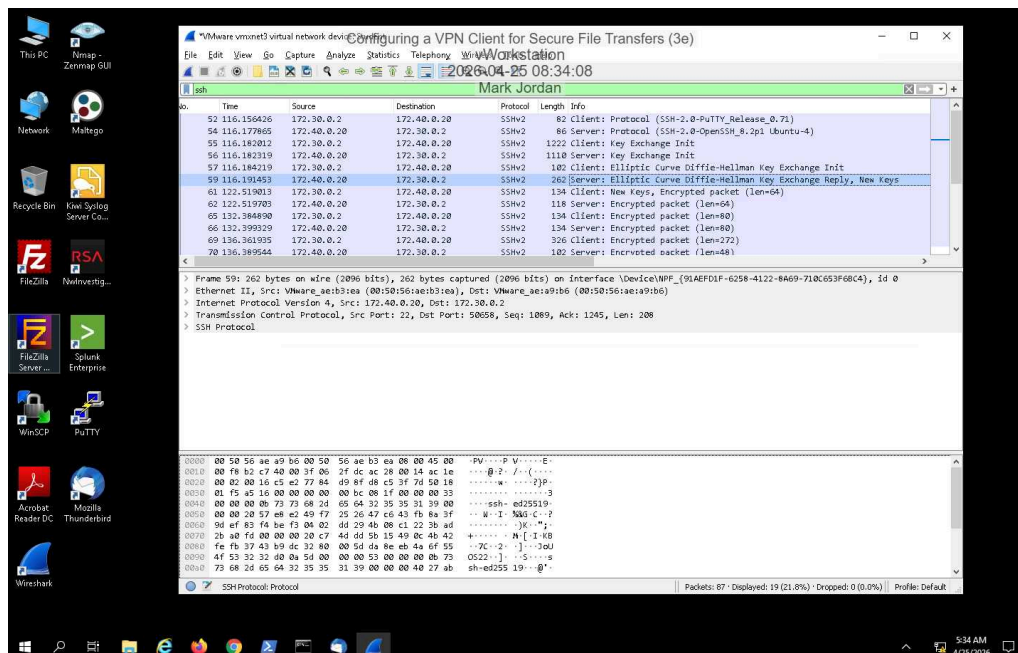
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24. Make a screen capture showing the contents of the file.txt file in the packet bytes pane.



27. Make a screen capture showing the filtered SSH packets in your capture file.



Section 3: Challenge and Analysis

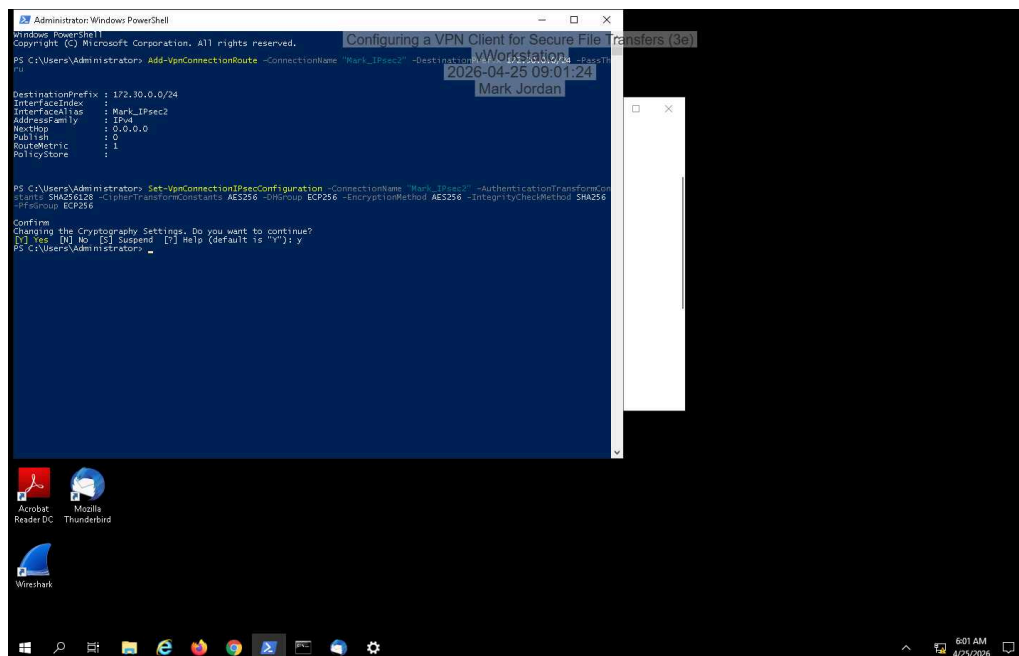
Part 1: Create a New VPN Connection using PowerShell

Document the command you used to add your VPN connection.

```
Add-VpnConnection -Name "Mark_IPsec2" -ServerAddress "202.20.1.2" -TunnelType IKEv2  
-EncryptionLevel Required -AuthenticationMethod EAP -SplitTunneling -AllUserConnection  
Add-VpnConnectionRoute -ConnectionName "Mark_IPsec2" -DestinationPrefix 172.30.0.0/24  
-PassThru
```

Part 2: Implement a Custom IPsec Policy

Make a screen capture showing the successfully executed `Set-VpnConnectionIPsecConfiguration` command in PowerShell.

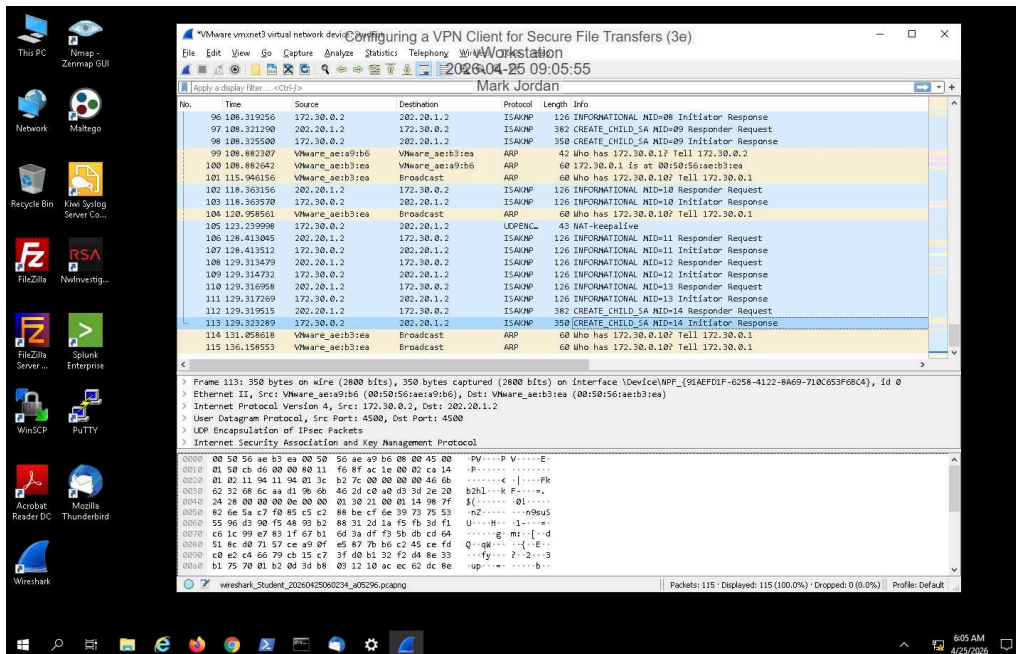


Part 3: Verify Your VPN Implementation using Wireshark

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Make a screen capture showing the **CREATE_CHILD_SA** exchange.



Make a screen capture showing the selected Diffie-Hellman transform.

